

Nanofluidic Detection Platform for Simultaneous Light Absorption and Scattering Measurement of Individual Nanoparticles in Flow

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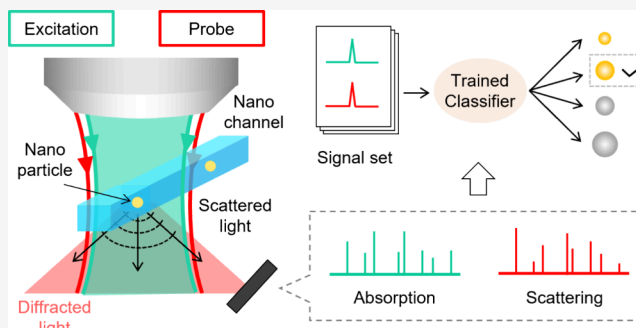


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ABSTRACT: Characterization and quantification of plasmonic nanoparticles at the single particle level have become increasingly important with the advancements in nanotechnology and their application to various biological analyses including diagnostics, photothermal therapy, and immunoassays. While various nanoparticle detection methodologies have been developed and widely used, simultaneous measurement of light absorption and scattering from individual plasmonic nanoparticles in flow is still challenging. Herein, we describe a novel nanofluidic detection platform that enables simultaneous measurement of absorption and scattering signals from individual nanoparticles within a nanochannel. Our detection platform utilized optical diffraction phenomena by a single nanochannel as both a readout signal for photothermal detection and a reference light for interferometric scattering detection. Through the elucidation of the frequency effect on the detection performance and optimization of experimental conditions, we achieved the classification of gold and silver nanoparticles with a diameter of 20–60 nm at an average accuracy score of $82.6 \pm 2.1\%$ by measured data sets of absorption and scattering signals. Furthermore, we demonstrated the concentration determination of plasmonic nanoparticle mixtures using a trained Support vector machine (SVM) classifier. Our simple yet sensitive nanofluidic detection platform will be a valuable tool for the analysis of nanoparticles and their applications to chemical and biological assays.



INTRODUCTION

Plasmonic nanoparticles have received significant attention due to their unique absorption and scattering characteristics.^{1,2} These properties offer a wide range of applications, including their use as sensing probes for in vivo imaging and in vitro diagnostics,^{3,4} local heat sources for photothermal therapy,⁵ and functional materials in chemical and biological assays.⁶ Consequently, accurate characterization of plasmonic nanoparticles is essential for both fundamental research and practical applications. In particular, sensitive detection at the single particle level is crucial for the identification and quantification of individual plasmonic nanoparticles.⁷ Single nanoparticle detection is also useful for the analyses of heterogeneous and complex samples, which is difficult with conventional bulk analysis measuring the average signal of an ensemble. In this respect, various microscopy-based single nanoparticle analysis methods have been developed. These methods include direct observation techniques such as scanning electron microscopy (SEM) and transmission electron microscopy (TEM).⁸ Additionally, sensitive scattering microscopy and photothermal absorption microscopy demonstrated the measurement of optical properties at the single particle level.^{9–12} However, such microscopy-based methods often require surface immobilization of the particles, which limits the throughput and observation in their native state in sample solutions.

Light scattering and absorption detection of individual nanoparticles in flow-based systems solves the above problems. One of the difficulties in light scattering detection of nanoscale objects in flow is low sensitivity due to short signal integration times and the rapid decay of the scattering intensity with particle diameter. Nevertheless, several techniques have been developed for the sensitive measurement of light scattering from individual nanoparticles within flow-based systems. Nanoparticle tracking analysis (NTA) enables concentration determination and size measurement of nanoparticles by suspended solutions.^{13,14} While NTA is typically employed for suspended solutions, flow-based NTA has also been developed and integrated with separation techniques.¹⁵ Flow cytometric nanoparticle detection methods utilizing specialized optical and fluidic systems have been developed for sensitive side-scattering detection of plasmonic nanoparticles and biological nanoparticles such as viruses and extracellular vesicles within a flow cell.^{16–18} Another promising approach for sensitivity improvement is interfero-

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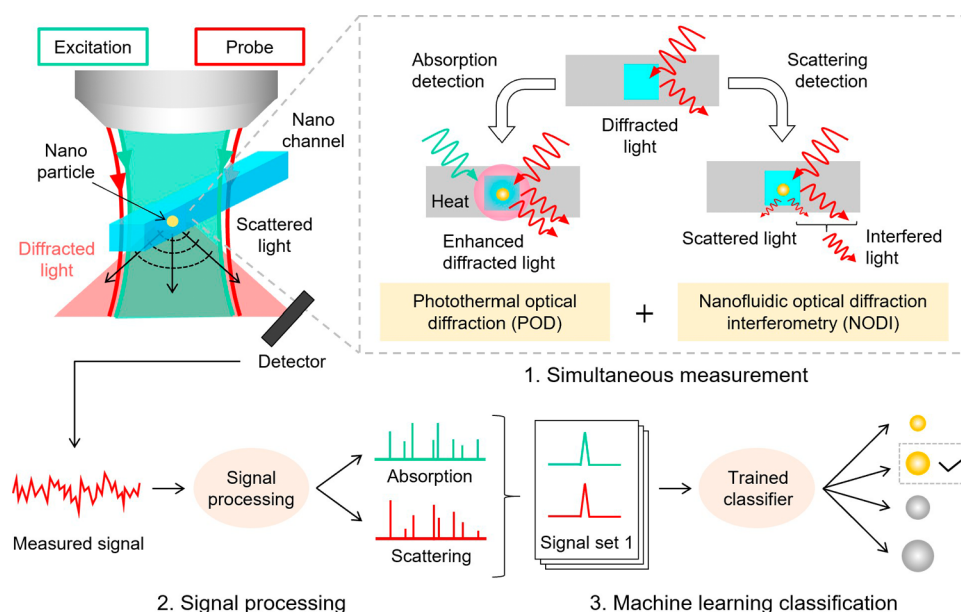


Figure 1. Schematic representation of our detection platform showing simultaneous light absorption and scattering measurement by POD and NODI, extraction of each component by signal processing, and signal-based machine learning classification.

metric scattering detection in which the interference of weak scattering light and relatively strong reference light enables the sensitive measurement of scattering properties.^{19,20} Several interferometric detection schemes have been developed for light scattering measurement of nanoparticles in flow such as dark-field illumination and heterodyne detection.^{21,22}

The sensitivity also becomes an issue for absorption detection of nanoscale objects in flow, and conventional absorbance measurements cannot be readily applied to single nanoparticle analysis. Photothermal detection methods utilizing light absorption and heat generation (photothermal effect) provide highly sensitive absorption measurement of target samples including nanoparticles in microfluidic devices.²³ For example, a thermal lens detection system combined with a hydrodynamic focusing system demonstrated 500 nm polystyrene bead measurement with high detection efficiency.²⁴ Furthermore, differential interferometric photothermal detection methods achieved highly sensitive nanoparticle detection in microchannels and nanochannels.^{25,26}

Although various techniques have been developed for measuring light absorption or scattering of individual nanoparticles, simultaneous measurement of light absorption and scattering in flow is still challenging despite the potential for achieving more accurate characterization and classification of plasmonic nanoparticles with unique optical properties. Microscopy-based nanoparticle analysis methods have achieved the measurement of light absorption and scattering for the same single nanoparticle immobilized on surfaces.²⁷ However, such a detection scheme is not applicable to flowing nanoparticles. A combination of measurement techniques developed for nanoparticle detection in flow as described above might seem a simple but promising approach. However, the complexity associated with achieving high sensitivity for nanoparticle detection constrains the parallelization, integration, and practical implementation of such detection systems.

Herein, we describe a novel nanofluidic detection platform that enables simultaneous measurement of absorption and scattering signals from individual nanoparticles in flow. Recently, we developed photothermal optical diffraction

(POD) detection method that utilizes optical diffraction from a single nanochannel as a readout for the photothermal effect of individual nanoparticles flowing within the nanochannel.^{28,29} Furthermore, we also developed nanofluidic optical diffraction interferometry (NODI) that utilizes optical diffraction from a single nanochannel as a reference light for interferometric detection of individual nanoparticles in flow.³⁰ The combination of these detection methods offers a promising approach for the simultaneous measurement of light absorption and scattering of nanoparticles through a simple experimental system and workflow. However, the similarity in detection principles, utilizing optical diffraction by a nanochannel, leads to difficulties in extracting and separating the POD signal and NODI signal independently with conventional signal processing systems. In this study, we established a nanofluidic detection platform that integrate these measurement methods and demonstrated accurate characterization and classification of individual nanoparticles.

First, we verified the principle of the proposed detection platform and demonstrated simultaneous measurement of light absorption and scattering from individual nanoparticles in flow. Next, we investigated the influence of modulation and extraction frequency, which are critical parameters affecting detection performance. After optimizing experimental conditions, we evaluated the capability of our detection platform for the characterization and classification of individual plasmonic nanoparticles with different absorption and scattering properties. Through simultaneous measurements of light absorption and scattering, we verified the classification of nanoparticle ensembles that cannot be differentiated based solely on either absorption or scattering signals. Finally, we demonstrated the mixing ratio estimation of plasmonic nanoparticle solutions from measured absorption and scattering signals and trained SVM-based classifier.

EXPERIMENTAL SECTION

Detection System and Workflow. Figure 1 shows the schematic representation of our detection platform. We simultaneously measured both absorption and scattering signals

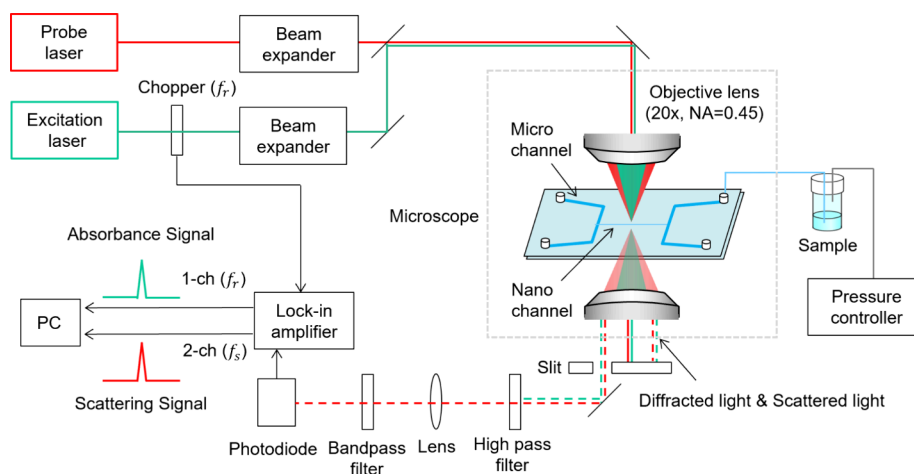


Figure 2. Experimental system for the simultaneous measurement of light absorption and scattering from individual nanoparticles flowing within a nanochannel.

from individual nanoparticles flowing within a nanochannel by the integration of two recently developed detection methods utilizing the optical diffraction phenomena of a single nanochannel: photothermal optical diffraction (POD) and nanofluidic optical diffraction interferometry (NODI). The detailed detection principle was described in our previous papers.^{29,30} Briefly, the POD measures the light absorption of individual nanoparticles by the photothermal effect. The diffracted light intensity depends on the refractive index difference between water and the glass substrate. The photothermal effect (light absorption and heat generation) of nanoparticles induces a local temperature increase and refractive index change, leading to a diffracted light intensity change. Thus, POD can measure the light absorption of individual nanoparticles through the diffracted light intensity. On the other hand, the NODI measures the light scattering of individual nanoparticles by optical interference. Although light scattering from nanoparticles is weak and difficult to detect without a sophisticated experimental setup, the interference of the scattering light from nanoparticles and the diffracted light from a single nanochannel enables sensitive measurement of the scattering properties of nanoparticles. One of the difficulties in the integration of POD and NODI for simultaneous absorption and scattering measurement is the separation and detection of signals arising from each measurement method. In this study, we established a frequency-based signal extraction and detection system utilizing a two-channel lock-in amplifier. Specifically, we extracted high-frequency components synchronized and nonsynchronized with the modulation frequency of the excitation laser to measure both the POD signal (synchronized with the modulation frequency) and the NODI signal (nonsynchronized with the modulation frequency) at the same time. This signal processing enables simultaneous measurement of light absorption and scattering from individual nanoparticles by a simple experimental system with two laser lines and a single photodetector. Additionally, a machine learning approach was employed to classify individual nanoparticles based on their absorption and scattering characteristics.

Optical Setup. Figure 2 shows the experimental setup of our detection platform. A 532 nm continuous-wave solid-state laser (Verdi-S M, Coherent Inc.) was used as an excitation beam for POD signal measurement. The intensity of the excitation beam was modulated by a light chopper (S584A, NF Corporation,

Japan). A 660 nm solid-state laser (gem660, Laser Quantum Ltd., England) was used as a probe beam for POD signal measurement and NODI signal measurement. The laser intensity under the objective lens was 10 mW for 532 nm and 1–3 mW for 660 nm lasers unless otherwise mentioned. These values were determined considering the optimal ratio of the absorption and scattering signal sensitivities for measured samples. Probe and excitation lasers were focused on a nanochannel by an objective lens (20 $\times$, NA = 0.45) equipped with a microscope. After passing through another objective lens (NA = 0.9), a mechanical slit (1 mm width), a high-pass filter to cut excitation laser, a focusing lens, and a band-pass filter to cut external light, diffracted light by the nanochannel and scattering light from nanoparticles were detected using a single photodiode with a 400 μ m aperture pinhole (ET-2030, Electro-Optics Technology Inc.). Light absorption and scattering signals were extracted by a two-channel lock-in amplifier (LI5660, NF Corporation, Japan). In the first channel (1-ch), the component synchronized to the modulation frequency was extracted to selectively detect the diffracted light intensity change by the excitation laser absorption of nanoparticles (POD signal). In the second channel (2-ch), the high-frequency component different from the modulation frequency was extracted to selectively detect the interferometric signal generated with the combination of diffracted light by a nanochannel and scattering light by a nanoparticle (NODI signal). The time constant of the lock-in amplifier was set to 1–2 ms, and measured signals were recorded by a data logger.

Fabrication of Nanofluidic Devices. Our nanofluidic devices consist of nanochannels for signal measurement and microchannels for sample injection to nanochannels. These channels were fabricated on a glass substrate (70 mm $\times$ 30 mm $\times$ 0.7 mm thickness, VIO-SIL, Shin-Etsu Quartz Co., Ltd., Tokyo, Japan). Briefly, nanochannels were fabricated by electron-beam lithography and dry etching, and microchannels were fabricated on another glass substrate by photolithography and dry etching. After drilling inlet and outlet holes, these substrates were cleaned and bonded together at 110 $^{\circ}$ C for 3 h. The size of fabricated nanochannels was 800–1200 nm wide and 550 nm deep, which were measured by a scanning electron lithography instrument (Regulus8230, Hitachi High-Technologies Corp., Japan) and stylus profiler (dektakXT, Bruker Corp.). The size of fabricated microchannels was 500 μ m wide and 4.5

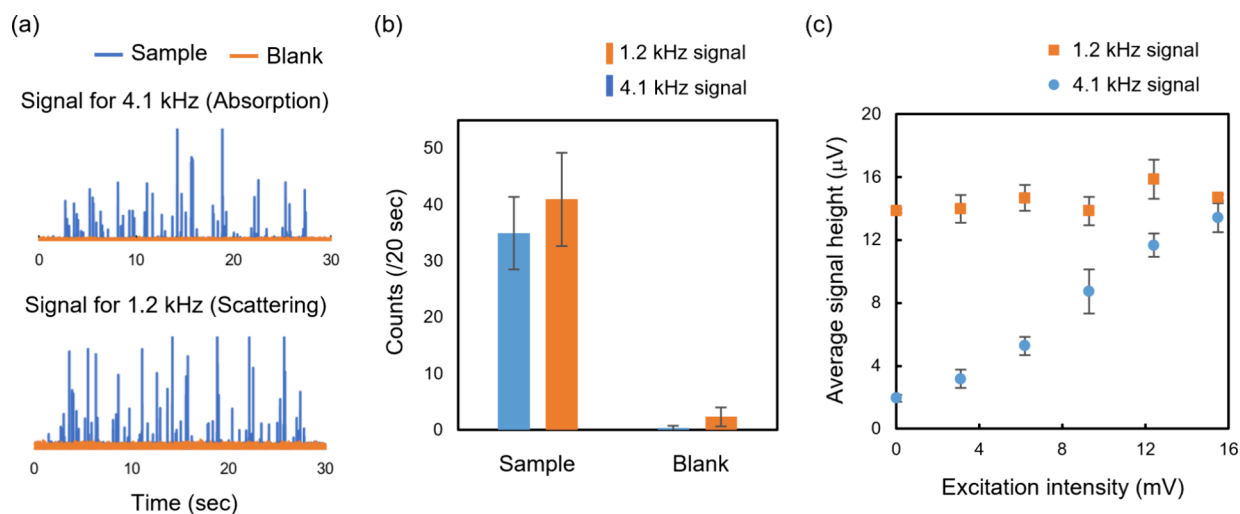


Figure 3. Verification of the simultaneous measurement system. (a) Example of signal components extracted at 4.1 kHz (synchronized with modulation frequency) and 1.2 kHz (nonsynchronized with modulation frequency). (b) Number of pulse signals measured and counted in 20 s. (c) Average signal height versus excitation intensity. For each condition, the average signal height of 100 pulse signals was calculated three times.

μm deep. The detailed fabrication procedure of nanofluidic devices is shown in a previous report.³¹

Fluidic System. Sample injection and replacement were performed by a home-built pressure-based fluidic control system shown in our previous report.³² Briefly, sample solutions were stored in vials and introduced into the nanofluidic devices by applying pressure to the vials, which were connected to microchannel inlets by capillaries (0.26 mm inner diameter, Institute of Microchemical Technology Co., Ltd., Japan). The flow rate within the nanochannel was regulated by a pressure controller (MFC5-EZ, Fluigent Inc., France). The actual flow rate was calculated from the replacement time of the sample solution in nanochannels. The typical flow rate in the nanochannel for 100 kPa pressure was 0.2 mm/s, and individual nanoparticles introduced into the nanochannel passed through the detection region within 10 ms duration.

Sample Preparation. Gold and silver nanoparticle solutions with 20–60 nm diameter were purchased from Sigma-Aldrich Co. LLC. Gold nanoparticle solutions (20 nm) were diluted to 100 pM (6.0×10^{10} particles/mL). Other nanoparticle solutions were used without dilution. The average number of nanoparticles in the laser focus calculated from the laser spot diameter and cross-section area of the nanochannel was <0.1 at prepared sample concentrations, suggesting that most of the measured signal was derived from a single nanoparticle. All solutions were filtered with a $0.22 \mu\text{m}$ syringe filter to avoid the clogging of nanochannels by nanoparticle agglomerates.

Data Processing and Analysis. Data processing and analysis were implemented using Python version 3.7.6 and the scikit-learn library (Python Software Foundation, Wilmington, DE, USA). First, pulse signals were extracted from recorded data in one of the two detection channels using the `Scipy.signal.find_peaks()` function. Pulse signal extraction thresholds were set at a level of background fluctuation $+5$ – 10σ with a minimum width of 2.5 ms. The minimum interval between adjacent pulses was set to 100 ms to avoid duplicate counting. Subsequently, the recorded data of the other channel was cut out in a time window of 50 ms before and after each extracted pulse signal, and the highest pulse signal within the time window was extracted as the corresponding second pulse

signal. Finally, the height and width of the identified pulse signal pairs were summarized for subsequent analysis. We employed a support vector machine (SVM) algorithm to classify individual nanoparticles using the prepared data sets. The radial basis function kernel was used for classification, with the cost parameter and kernel parameter set to 1.0 and 0.1. These hyperparameters were determined through a grid search process with 10-fold cross-validations and a comparison of average accuracy ratios.

RESULTS

Verification of the Simultaneous Measurement System. First, we verified the principle of our detection platform. Figure 3a shows the signals for 40 nm gold nanoparticles measured by our detection system. In this experiment, the modulation frequency of the excitation laser was set to 4.1 kHz, and we measured the POD signal and the NODI signal by extracting the signal components of 4.1 kHz (1-ch) and 1.2 kHz (2-ch), respectively. Pulse signals above the background fluctuation were observed for both frequency components. Figure 3b shows the number of pulse signals counted based on the threshold set at the background fluctuation $+5\sigma$. These results suggest that the POD signal and the NODI signal from individual nanoparticles were successfully detected. Subsequently, we investigated signal dependency on probe and excitation lasers to elucidate the signal generation mechanism for each channel. Experimental results show that 1.2 kHz component pulse signals were related to the probe laser rather than the excitation laser, while 4.1 kHz component pulse signals were related to both the probe and excitation laser (Figure S1). Furthermore, we investigated the relationship between signal intensity and excitation laser power. We used recorded data for the second channel and a threshold set at background fluctuation $+10\sigma$ to extract pulse signals and their positions. As shown in Figure 3c and Figure S2, the average intensity of 1.2 kHz component pulse signals was almost constant for different excitation laser intensities, while the average intensity of 4.1 kHz component pulse signals was proportional to the excitation laser intensity. These results show that measured pulse signals for 1.2 and 4.1 kHz components were derived from different signal generation mechanisms, and the trends were in good agreement

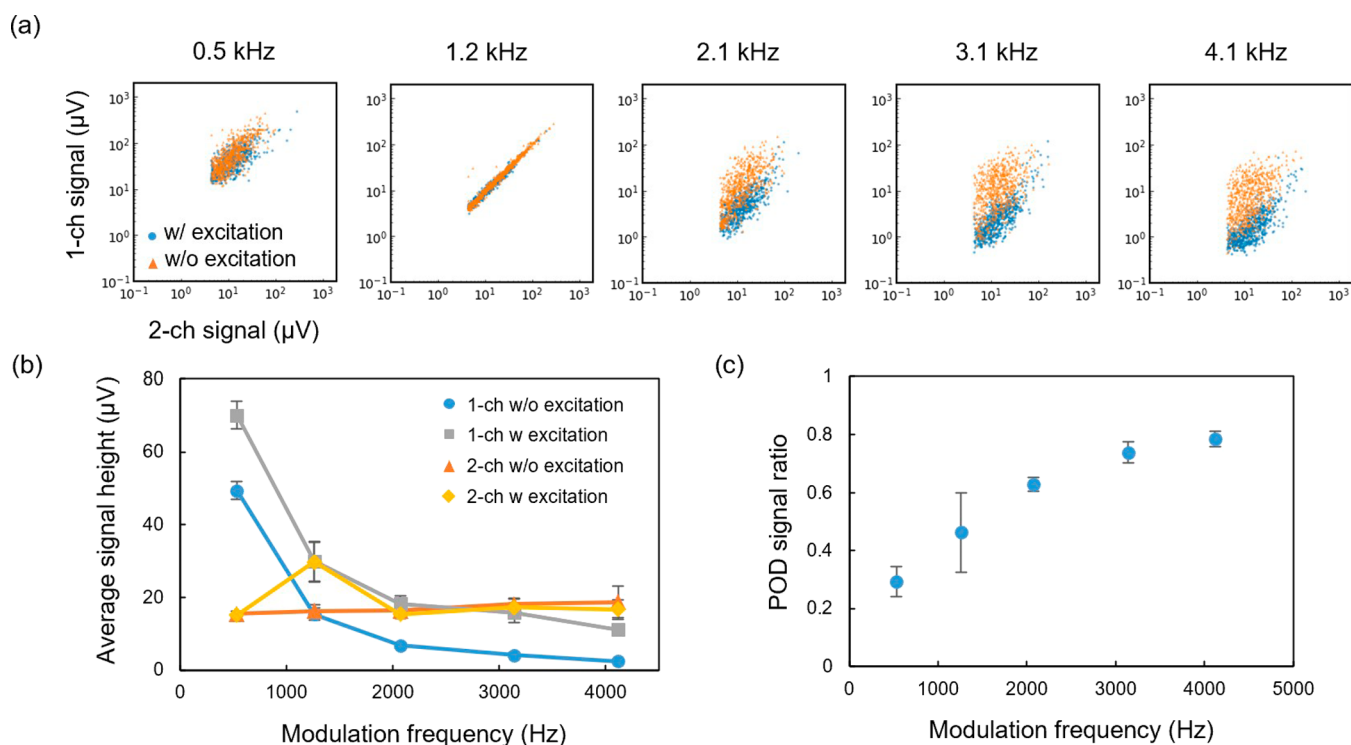


Figure 4. Effect of the modulation frequency of the excitation laser on the detection performance. (a) Scatter plots of signal heights, (b) average signal height, and (c) photothermal signal ratio for different modulation and extraction frequency conditions. For each condition, 600 data sets of signal heights were plotted, and the average signal height of 200 pulse signals was calculated three times.

with the expected results for the proposed detection principle for POD and NODI. Thus, our detection platform realized simultaneous absorption and scattering measurements of individual nanoparticles in flow.

Effect of Modulation and Extraction Frequency. While the proposed detection platform enables simultaneous absorption and scattering measurement of individual nanoparticles by a simple experimental system with two lasers and a single photodetector, such a measurement system requires separation and detection of two different signals without crosstalk. In our detection platform, absorption and scattering signals were separately measured by extracting different frequency components using a lock-in amplifier; modulation and extraction frequencies play a crucial role in ensuring the reliability of these measurements. Therefore, we investigated the effect of the modulation and extraction frequency on the detection performance. Figure 4a shows the scattering plots of measured signals for different frequency conditions. In this experiment, we changed the modulation frequency of the excitation laser from 0.5 kHz to 4.1 kHz and measured the signal in the first channel at the same extraction frequency as the modulation frequency. The extraction frequency for the second channel was set to 1.2 kHz for all experiments since the highest signal-to-noise ratio can be obtained around this frequency as shown in our previous experiments. We used recorded data for the second channel and a threshold set at background fluctuation $+10\sigma$ to extract pulse signals and evaluate the performance. Examples of measured pulse signals and scatter plots are summarized in Figure S3 and Figure 4a. For the modulation frequency of 0.5 kHz, there was some variation in signal intensity, but little difference in the presence or absence of the excitation laser. These results imply that the high-intensity NODI signal leaked into the first channel, impeding the accurate extraction of the POD signal. For the

modulation frequency of 1.2 kHz, the leakage and overlapping of the scattering signal and the photothermal signal occurred due to the proximity of the extraction frequencies, leading to a line-shaped scatter plot. In contrast, the results for the modulation frequency of 2.1 kHz exhibit a difference in signal intensity for the first channel in the presence or absence of the excitation laser, suggesting a partial separation between the POD signal and the NODI signal. The difference further increased for the modulation frequency of 3.1 and 4.1 kHz. The average signal height for each experimental condition is summarized in Figure 4b. Except for the 1.2 kHz modulation frequency, the average signal height for the second channel remained nearly constant in intensity regardless of the presence or absence of the excitation laser across the measured range of modulation frequencies. These results indicate that selective measurement of the NODI signal is feasible by assigning an extraction frequency different from the modulation frequency of the excitation laser. On the contrary, the average signal height for the first channel exhibited significant variations depending on the modulation frequency. One of the reasons for this dependency originates from the frequency-dependent nature of the POD signal. As shown in the previous literature, the POD signal intensity varies inversely with the modulation frequency. Another reason is the leakage of the NODI signal as described above. Our previous results indicate an inverse relationship between the NODI signal and the extraction frequency. As indicated by the blue line in Figure 4b, the average signal height for low modulation and extraction frequency remained high even without the excitation laser, suggesting the measurement of both the POD signal and the NODI signal. Therefore, to selectively measure the POD signal, it is necessary to consider these factors and set the modulation frequency accordingly. Figure 4c shows the ratio of the POD signal to the pulse signal intensity measured in the first channel.

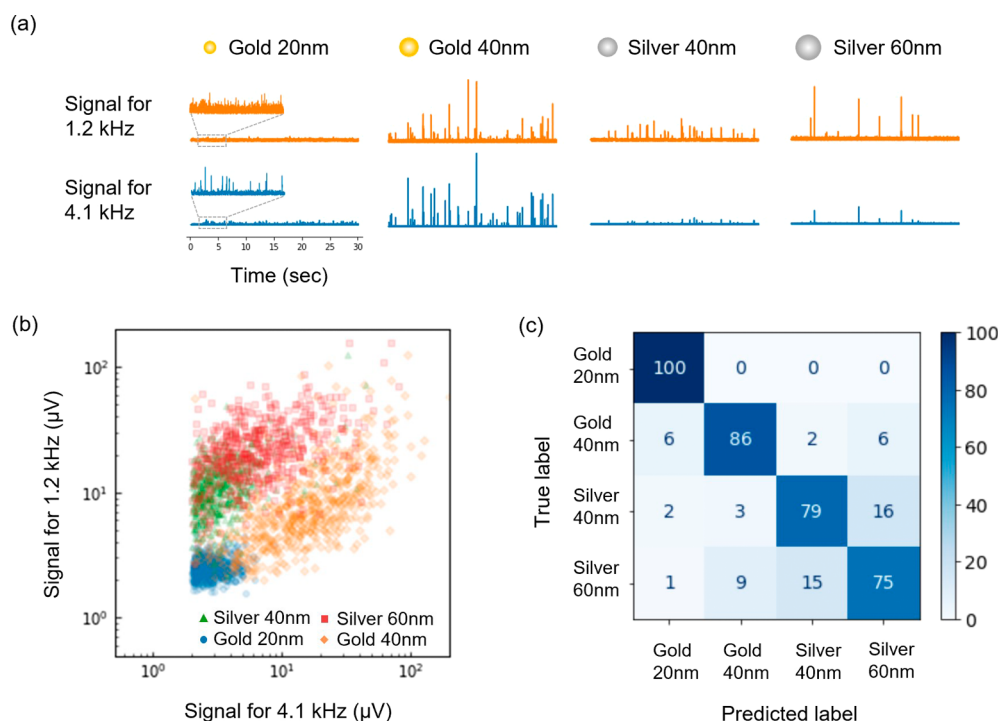


Figure 5. Classification of individual plasmonic nanoparticles with different absorption and scattering spectra. (a) Example of measured signals for 20–60 nm gold and silver nanoparticle solutions. (b) Scatter plot of signal heights for 600 data sets of each sample. (c) Confusion matrix for the classification of four types of nanoparticles using the support vector machine (SVM).

The POD signal intensity was calculated from the difference in measured signal intensity with and without the excitation laser. The contribution of the POD signal in the measured signal intensity exceeded 80% at a modulation and extraction frequency of 4.1 kHz. From these results, we concluded that high modulation and extraction frequency enable the measurement of the POD signal while suppressing the effect of the NODI signal leakage. Note that the influence of signal leakage becomes significant for nanoparticles with high scattering efficiency and low absorption efficiency, or vice versa. In such cases, accurate characterization of scattering and absorption efficiency directly from measured signals becomes challenging, and computational corrections by estimating the effect of leakage are necessary to mitigate this issue.³³

Classification of Nanoparticles Based on Absorption and Scattering Signals. Plasmonic metallic nanoparticles exhibit unique absorption and scattering spectra, which are influenced by factors such as their size, shape, and composition.^{2,34} Therefore, simultaneous measurement of absorption and scattering enables precise characterization and classification of plasmonic nanoparticles at a single particle resolution. In this section, we demonstrate the capability of our detection platform for the measurement and classification of individual plasmonic nanoparticles with different absorption and scattering spectra. Figure 5a shows the measured signals for four types of plasmonic nanoparticles (20 nm gold, 40 nm gold, 40 nm silver, and 60 nm silver nanoparticles). The modulation frequency of the excitation laser was set to 4.1 kHz, and extraction frequencies for the first and second channels were set to 4.1 and 1.2 kHz. Experimental results show that pulse signals with different heights were observed for each sample, except for 20 nm gold nanoparticles where the signal height for 1.2 kHz was lower than the background fluctuation. We used recorded data for the first channel and a threshold set at 2 μV (background

fluctuation +7–8σ) to extract pulse signals and their positions. Figure 5b shows the scatter plot of signal heights of the 600 data sets for each sample. Four types of nanoparticles were plotted with a slight overlap but in different positions, suggesting that nanoparticles can be classified by our detection platform. To quantitatively evaluate the classification performance, we employed SVM classification of individual nanoparticles using the obtained data sets. The width and height of signals for both detection channels were used as feature parameters. For each sample, 500 data sets were used as training data, and 100 data sets were used as test data. Figure 5c shows the confusion matrix for the classification of four types of nanoparticles. Although there was some degree of misclassification among nanoparticles with similar sizes or the same materials, our detection platform achieved accurate discrimination of the four nanoparticle types. The average accuracy score for 10-fold cross-validation was $82.6 \pm 2.1\%$ at the optimized hyperparameter condition. Also, the average accuracy score for the classification of two types of nanoparticles exceeded 90% except for the classification of 40 and 60 nm silver nanoparticles (Figure S4). Furthermore, we assessed the improvement in classification performance by simultaneous light absorption and scattering measurements. Figure S5 shows the confusion matrix when only one of the signals for 4.1 and 1.2 kHz was used for classification. In both cases, classification performance was significantly worse for certain pairs of nanoparticles, and the average accuracy score for 10-fold cross-validation decreased to 55.7 ± 2.5 and $66.5 \pm 2.8\%$, respectively. These results demonstrated the effectiveness of simultaneous absorption and scattering measurements by POD and NODI for accurate nanoparticle classification. Note that not all nanoparticles exhibited detectable scattering and absorption signals in this experiment. For instance, the absorption signal of 20 nm gold nanoparticles was relatively weak, making it difficult to distinguish signals from background

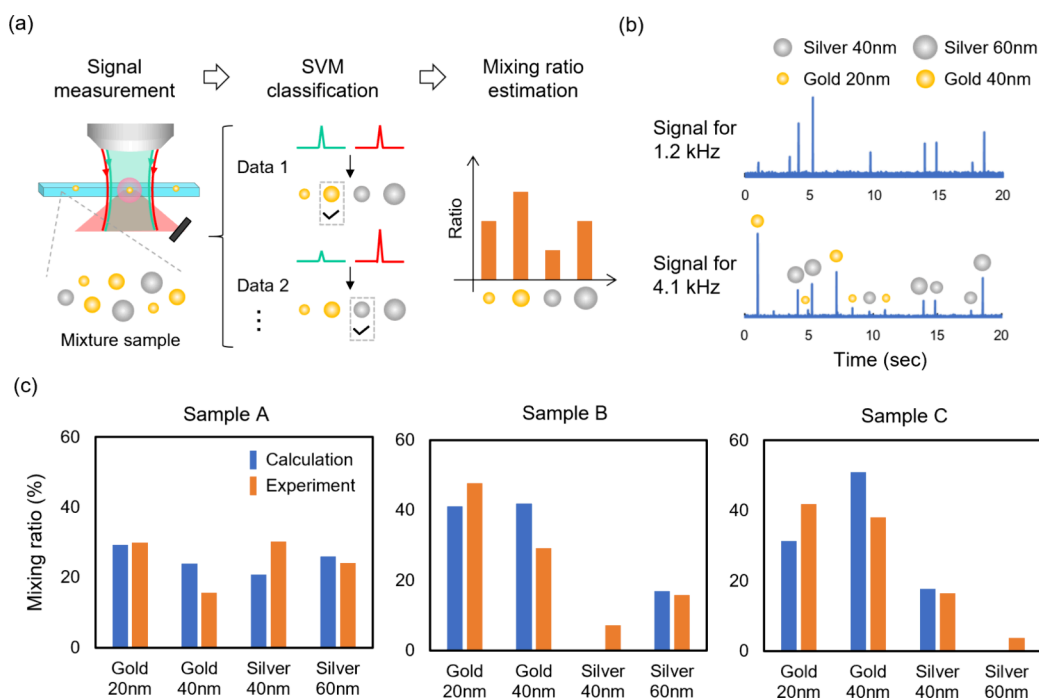


Figure 6. Demonstration of estimating sample mixing ratio by our detection platform. (a) Experimental workflow of mixing ratio estimation. (b) Example of measured signals and predicted nanoparticles by a trained SVM classifier. (c) Results of mixing ratio estimation for three sample solutions. The mixing ratio was calculated from 1000 data sets of pulse signals for each sample.

fluctuations. Moreover, due to the high scattering efficiency and low absorption efficiency of silver nanoparticles at the measured wavelength,³⁴ the signal measured in 1-ch for 40 nm silver nanoparticles was mainly derived from not light absorption but light scattering. Further improvement in classification performance can be achieved through enhancements in sensitivity and optimization of experimental conditions.

Demonstration of Mixing Ratio Estimation. Detection and classification of nanoparticles at the single particle level enable the accurate quantification of mixed or heterogeneous samples, which is challenging to achieve through bulk analyses measuring the average signal of the ensemble. In this section, we demonstrate the capability of our detection platform to estimate mixing ratios of nanoparticle solutions. The workflow is illustrated in Figure 6a. First, we prepared sample solutions containing various nanoparticles and introduced them into the nanochannel to obtain absorption and scattering signals. Subsequently, we utilized a trained SVM classifier to predict the corresponding nanoparticles for each set of measured signal data. Finally, we estimated the mixing ratio by counting the assigned number for each nanoparticle and dividing them by the total number. In this experiment, we prepared three sample solutions containing four types of nanoparticles at different mixing ratios. Figure 6b shows the typical example of measured signals for prepared sample solutions. The types of nanoparticles estimated by the trained SVM classifier for each set of measured pulse signals are also shown. Figure S6 summarizes the prediction results of all measured signals for the three sample solutions, which are in good agreement with the results for each sample solution shown in Figure 5b. Figure 6c shows the results of mixing ratio estimation for three sample solutions. The experimentally estimated ratio was in broad agreement with the corresponding calculated ratio for all three samples. Furthermore, we estimated mixing ratios based on the classification probabilities. Using a trained SVM classifier, we obtained the

probability that the measured signal corresponds to which of the four types of nanoparticles for each data set. Mixing ratios were then calculated from the average probabilities of all data sets. This procedure achieved a more accurate estimation of the mixing ratios, as shown in Figure S7. These results demonstrated that our detection platform can be used for the analyses of mixed or heterogeneous nanoparticle samples by simultaneously measuring absorption and scattering signals at the single particle level.

DISCUSSION

To the best of our knowledge, no other method has achieved simultaneous measurement of absorption and scattering of individual nanoparticles in flow. Our detection platform demonstrated the feasibility of such simultaneous detection and signal-based classification of individual nanoparticles, while there is still room for improvement in the detection and classification performance. First, the improvement of detection sensitivity and accuracy will lead to more precise characterization and classification of nanoparticles. Sensitivity can be improved by increasing laser intensity, although this might be constrained by the optical trapping of nanoparticles.³⁵ Another practical approach is incorporating lasers with different wavelengths depending on the measurement target. The variability in the signal intensity depends not only on the particle itself but also on other factors such as the particle positions within the nanochannel, which is especially noticeable in interferometric scattering detection.¹⁹ In our detection system, the effect of positional signal variation can be somewhat mitigated by the Brownian motion of nanoparticles. An individual nanoparticle passes through the detection region in about 10 ms, during which the average diffusion length of the nanoparticles is comparable to or greater than the channel size. Consequently, the interference scattering signal from the nanoparticle fluctuates within a certain range during the

measurement due to different interferometric conditions, but the maximum signal value contains information about the particle's size and composition. Utilizing smaller nanochannels and employing appropriate beam shapes can further suppress such system-induced signal variations, leading to more accurate measurement and classification. The signal processing system and procedure are another important factor. As described above, our detection platform achieved the separation of the POD signal and the NODI signal through frequency component extraction by a lock-in amplifier. The effects of signal leakage can be reduced by setting the detection sensitivity of each measurement method appropriately. Furthermore, experimental results demonstrated that a higher modulation frequency of the excitation laser is effective in separately extracting these signals. Although the modulation frequency was kept below 4.1 kHz in this study due to equipment limitations, the effects of signal leakage can be further suppressed by modulating an excitation laser at a higher frequency using an acousto-optic modulator. In this study, simultaneous measurements were performed using a simple optical setup with two lasers and a single detector. In-line measurement of absorption and scattering using separate lasers and detectors enables more accurate measurements with less signal leakage, while this approach would make the setup more complex. For the machine learning classification, only the height and width of pulse signals for each channel were used as feature parameters for nanoparticle prediction. Additional feature parameters such as area, onset angle, and prominence of pulse signals might improve classification performance.

The capability of our detection platform to realize simultaneous measurement of absorption and scattering by a simple optical setup offers great potential for a wide range of nanoparticle analyses and applications. For example, our detection platform is not only applicable to the quantification and evaluation of plasmonic nanoparticles but also suitable for various biological assays using plasmonic nanoparticles or their aggregates as detection probes.^{36,37} Furthermore, by enhancing sensitivity and incorporating UV wavelengths, this method could be extended to applications involving other biological nanoparticles, such as viruses and extracellular vesicles.³⁸ Additionally, our detection platform can be combined with other detection methods such as fluorescence, allowing for more comprehensive evaluation and classification based on multi-dimensional information. Furthermore, the capability of measuring individual nanoparticles in flow enables the integration of our detection platform with nanofluidic separation and sorting techniques,^{39–41} offering in-depth analyses of various nanoparticles.

CONCLUSIONS

In this study, we developed a novel detection platform for simultaneous measurement of absorption and scattering signals from individual nanoparticles utilizing nanofluidic optical diffraction phenomena. Through the elucidation of the frequency effect on the detection performance and optimizing experimental conditions, our detection platform achieved the classification of individual plasmonic nanoparticles based on the measured data sets of absorption and scattering signals. We also demonstrated the mixing ratio estimation for nanoparticle mixtures using a trained SVM classifier. The developed simple and versatile detection methodology will contribute to the advancement of various nanoparticle analyses and applications, including chemical and biological assays utilizing plasmonic

nanoparticles, environmental monitoring, and clinical diagnostics targeting biological particles.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.analchem.4c01671>.

Investigation of signal dependency on probe and excitation lasers; example of measured signals for different experimental conditions; accuracy score for support vector machine classification; confusion matrix for the classification by absorbance or scattering signals; scatter plots for mixed sample solutions and signal-based prediction; mixing ratio prediction based on the classification probabilities (PDF)

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Notes

The authors declare no competing financial interest.

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